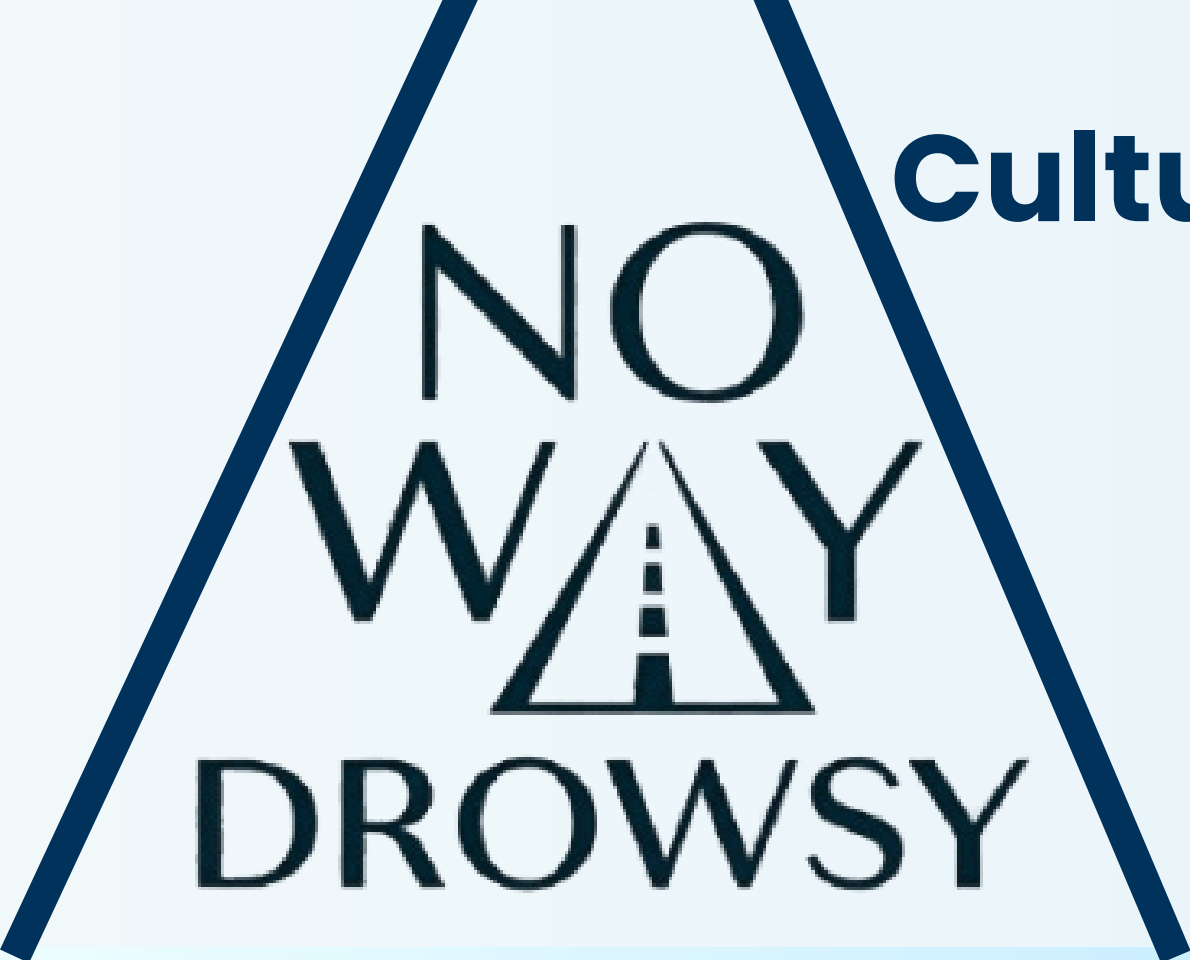




Culturally Robust Driver Drowsiness Detection

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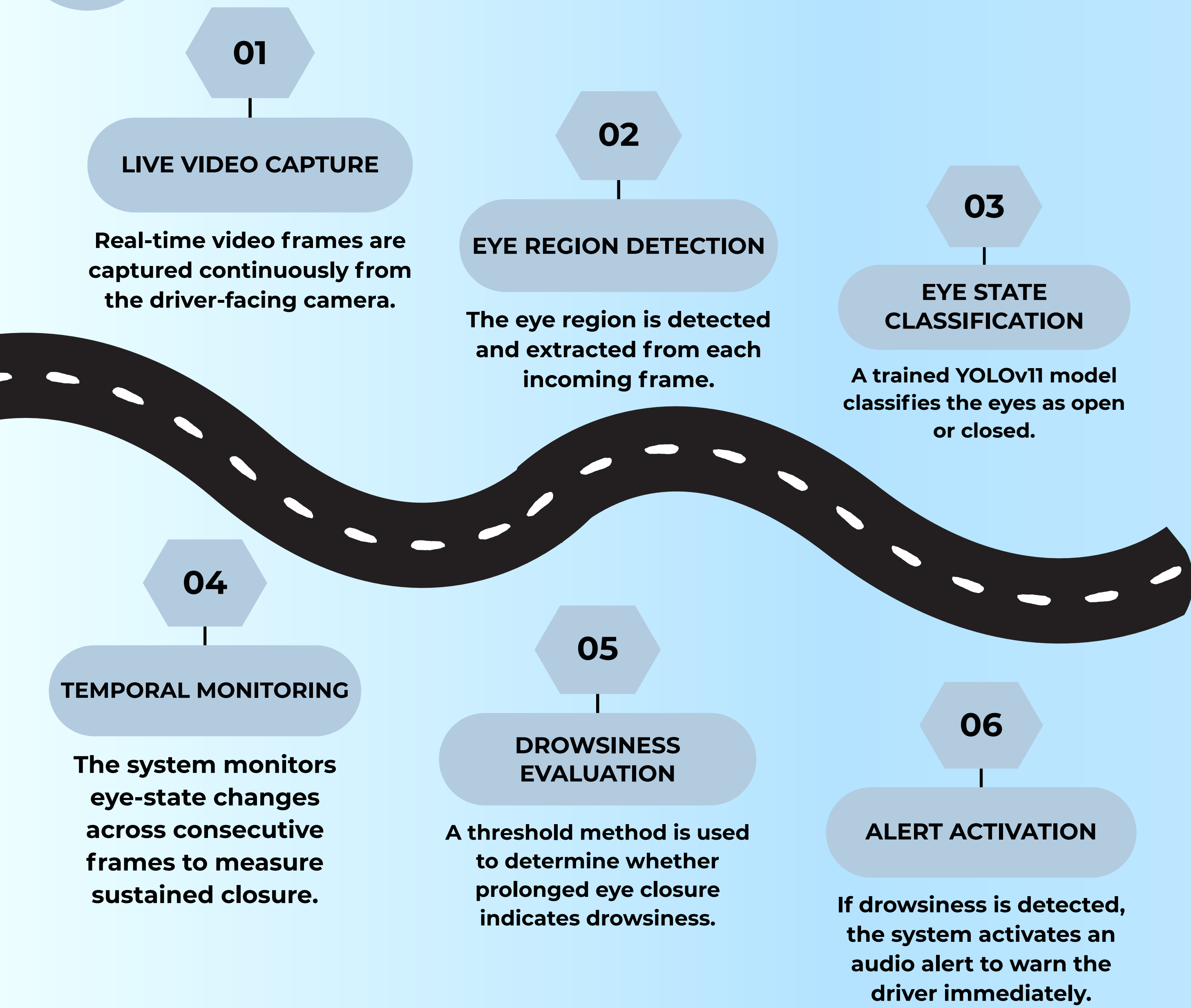


Introduction and Problem Specification

Drowsy driving is a major cause of road accidents because it reduces driver attention and slows reaction time. However, many existing driver-monitoring systems rely on full facial visibility or expensive hardware, making them less practical in Saudi Arabia and GCC countries, where drivers may wear niqab, hijab, shemagh, or masks. This project addresses the need for a practical and culturally adaptive drowsiness detection system. The project proposes an AI-based solution using a YOLO-based computer vision model and a mobile camera to detect visible cues such as eye closure and head movement. The system runs directly on the mobile device to preserve privacy and provides real-time audio alerts to improve driver safety.



Methodology



Dataset collection and Preprocessing

1. Balanced Hybrid Dataset (5,000 Images)

To ensure real-world robustness without model overfitting, a 4:1 ratio of baseline to custom data was established:

- **Public Baseline (80%):** 4,000 images sampled from a highly optimized open-source dataset to train foundational eye morphology, skin tones, and standard lighting variations.
- **Custom Edge-Cases (20%):** 1,000+ hand-annotated, locally sourced images specifically introducing severe cultural and environmental occlusions: the niqab, shemagh, glasses, and lens glare.

2. Annotation & Quality Assurance

- **Targeted Bounding Boxes:** Instead of full-face mapping, annotations were strictly isolated to the eye region, forcing the YOLOv11 model to ignore lower-face occlusions entirely.
- **"Pixel Reality" Threshold:** A strict binary classification (open/close) was enforced. If any portion of the iris or sclera remained visible, it was labeled open to virtually eliminate premature false alarms.
- **Unified Validation:** Public and custom data were merged prior to the train/val split. This ensured the model's validation metrics accurately reflected its performance on both standard faces and cultural edge cases simultaneously.



Results

Training Behavior:

- Training and validation losses decreased steadily across all epochs
- No signs of overfitting : training and validation curves remained closely aligned
- Model generalized well to unseen data

Key Findings:

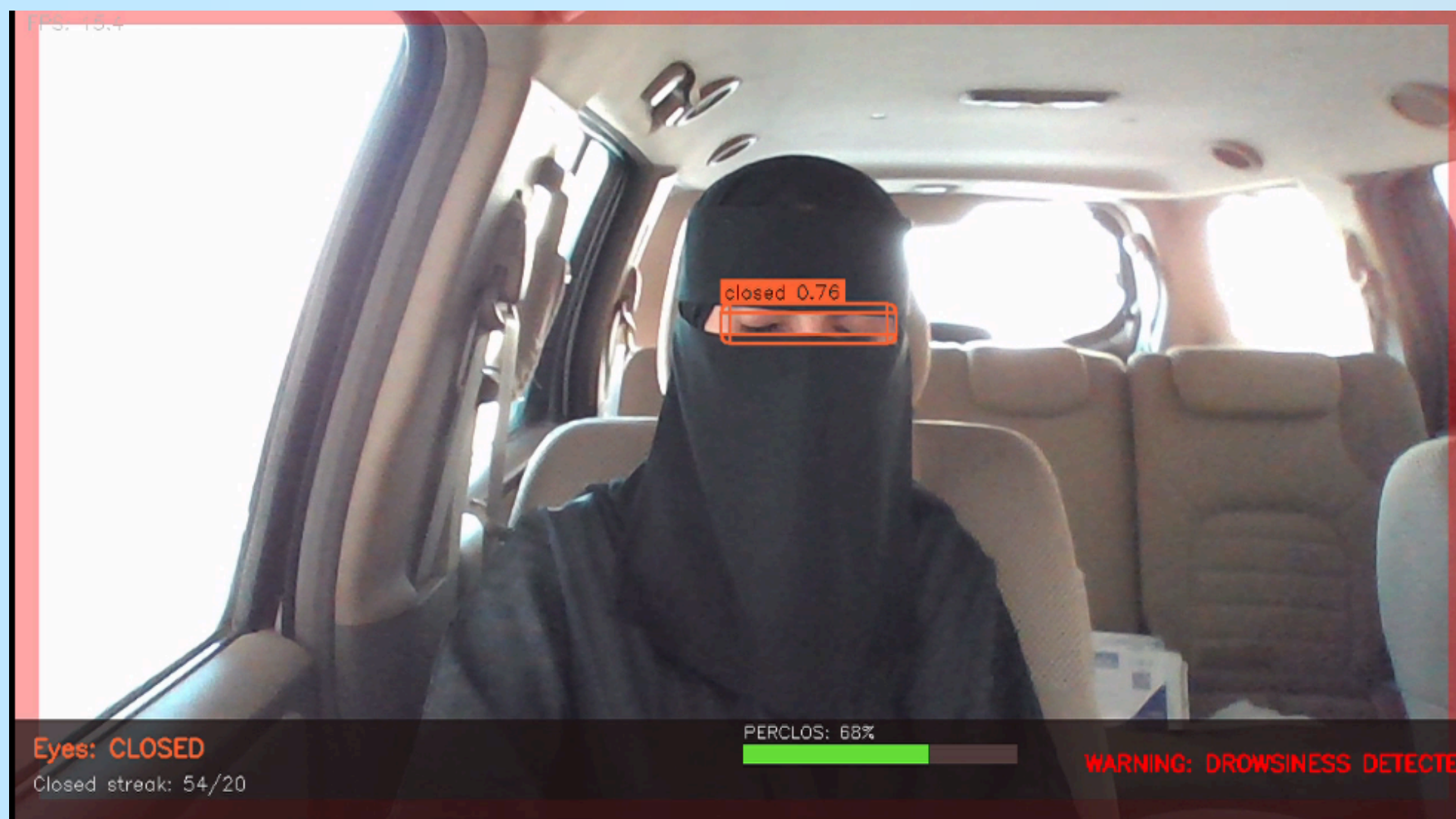
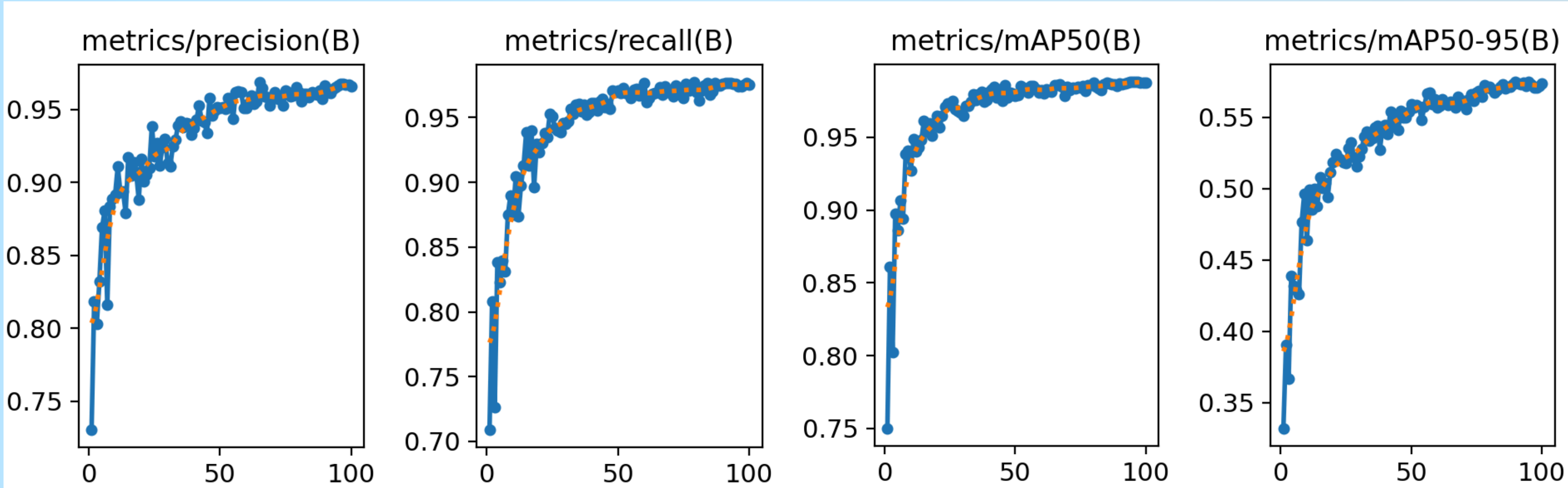
- **Robustness:** High accuracy maintained across diverse cultural and environmental scenarios.
- **Reliability:** The YOLO-based architecture provides a lightweight yet powerful solution for proactive road safety.

96%
PRECISION

97%
RECALL

98%
MAP@0.5

58%
MAP@0.5-95



Conclusion

Drowsy driving remains a critical safety risk, particularly in regions where face coverings limit the effectiveness of standard detection systems. This project delivers a YOLO-based, real-time drowsiness detector achieving 96% precision, 97% recall, and 98% mAP@0.5 trained on a culturally diverse dataset including niqab, shemagh, and occluded faces. Running entirely on-device, the system preserves privacy while issuing instant audio alerts, offering a practical and inclusive solution for driver safety in the GCC region.